

I. Query-Level Entropy Gating

Input image X_v (AMBER)



TEXT PROMPT X_t

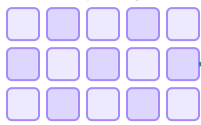
"Is there direct contact between the orange and the laptop?"

Visual Inertia



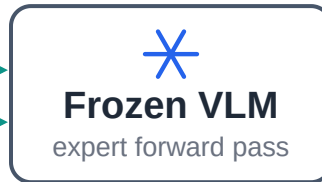
attention stays anchored on salient regions; grounding decays, priors dominate

(X_v, X_t)



visual tokens X_v

instruction tokens X_t



softmax

First-token distribution

P_{std} — with vision

Yes

No

near-uniform $\rightarrow H(Y_1)$ high

$H(Y_1) \geq \tau$?

Shannon entropy of first token

$H < \tau$ ✓

System 1 · Standard

$H < \tau$ · greedy AR decoding gate stands down

Y_{std} (confident queries)



Yes — vision-text conflict
route to System 2

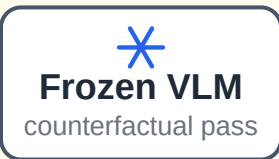
RESPONSE (GREEDY DECODING)

✓ **"Yes" — direct contact**

II. System 2 · Counterfactual Contrastive Decoding



text-only input — X_v omitted



Language prior z_{cf}

(text-only logits)

No

Yes

conservative prior favors "No"

$$z_{CD} = (1+\alpha) \cdot z_{std} - \alpha \cdot z_{cf}$$

$\forall y \in V_{head}$ (APC)

Calibrated $P(y)$

softmax over z_{CD}

Yes

No

$\alpha = 1.0 \cdot \beta = 0.1$

Frozen VLM visual tokens text tokens omitted vision

expert (with vision)

language prior (text-only)

calibrated (final)

confident path (System 1)

contrastive route (System 2)

$\tau \in [0.570, 0.700]$ · tuned per model